

KARTHIK SENGUPTA

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Education

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Master of Science in Computer Science | **GPA: 4.0/4.0**

Sep 2025 – Dec 2027

• **Coursework:** Algorithms, Database Management Systems, Program Design Paradigms, Pattern Recognition & Computer Vision

Vellore Institute of Technology

Vellore, India

Bachelor of Technology in Computer Science and Engineering | **GPA: 3.55/4.0**

Sep 2021 – May 2025

• **Coursework:** Artificial Intelligence, Natural Language Processing, Statistics, Data Structures & Algorithms, Cloud Computing

Skills

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, Vision Transformers (ViT), NumPy, Pandas, CUDA

Computer Vision: OpenCV, MediaPipe, image classification, real-time video processing, pose estimation

Generative AI & NLP: Large Language Models (LLMs), LLM fine-tuning, prompt engineering, RAG, LangChain, Hugging Face, Ollama, OpenAI API

Systems & Data: C++ (Modern C++), ML compute-graph compilation, SQL, MySQL, PostgreSQL, ETL, Star Schema, Data Warehousing, Power BI

Languages & Tools: Python, C++, Java, R, JavaScript; Git, Linux, Docker, CMake, AWS, Flask, REST APIs, GoogleTest, Agile/Scrum

Professional Experience

Research Intern

Jun 2024 – Jul 2024

National Center for Assistive Health Technology, IIT Delhi

New Delhi, India

- Built an AI-powered web application integrating the **ChatGPT API (LLM)** and **OpenCV** to automate real-time image description for 200+ visually impaired students across 8 education centers, achieving 82% accuracy validated by specialists.
- Engineered **Python NLP pipelines** for preprocessing, summarization, and descriptive analytics, cutting manual effort by 65%; deployed as **Flask REST APIs** with concurrent multi-center access and production-grade error handling.

Software Development Engineer Intern

Aug 2023 – Oct 2023

LTIMindtree

Chennai, India

- Built an HR analytics and automation platform on Microsoft Power Platform automating 500+ workflow events and cutting onboarding cycle time 35%; adoption grew 30% in daily active users within the first quarter.
- Designed **Power BI** dashboards and optimized **SQL** ingestion pipelines delivering real-time metrics to technical and non-technical stakeholders, eliminating 15+ hours/month of manual reporting.

Projects

minicompiler: C++ ML Compiler | *Modern C++, Eigen, CUDA, Linux, GoogleTest*

- Built a custom **C++ compiler** for **ML compute graphs** with a DAG-based intermediate representation and three optimization passes (dead-node elimination, constant folding, operator fusion) targeting an **Eigen** CPU backend.
- Benchmarked execution against **PyTorch**, visualized the IR with Graphviz, and maintained the codebase with automated **GoogleTest** suites on **Linux**.

FitFight: Pose-Controlled Fitness Fighting Game | *Python, MediaPipe BlazePose, OpenCV, PyTorch*

- Built a real-time **computer vision** system classifying 5 exercises from 33 BlazePose landmarks using joint-angle state machines with hysteresis and temporal smoothing, at 95%+ accuracy and 30+ FPS on CPU.
- Architected a decoupled, fully unit-tested game engine (zero rendering I/O) with progressive difficulty across 3 bosses and post-session performance tracking.

Document Retrieval & Summarization System | *Python, LangChain, OpenAI, ChromaDB, RAG*

GitHub

- Built a hybrid **RAG** pipeline over **large language models**, combining dense embeddings with BM25 retrieval and cross-encoder re-ranking to improve precision by 15% and cut response latency 40% across a 500-query benchmark.

Streaming Analytics Data Warehouse | *R, MySQL, SQLite, Star Schema, ETL*

- Built an **ETL pipeline** and Kimball **star-schema** warehouse ingesting 98,000+ transactions into a cloud-hosted MySQL datamart; authored analytical **SQL** (window functions, CTEs) to surface year-over-year growth trends.

Leadership & Publications

Publication: “RetailGPT: A Fine-Tuned LLM Architecture for Customer Experience and Sales Optimization,” *ICSSAS 2024* — fine-tuning methodology and deployment architecture for production LLM services.

President, Khoury Masters Student Council (Apr 2026 – Present): Lead governance and faculty relations for 150+ MS students; drafted the founding constitution and transitioned the council to a formal election model.

Treasurer, Khoury Masters Student Council (Dec 2025 – Mar 2026): Managed budgeting and coordinated 5+ professional and community events per semester with Khoury Operations and Recruitment teams.